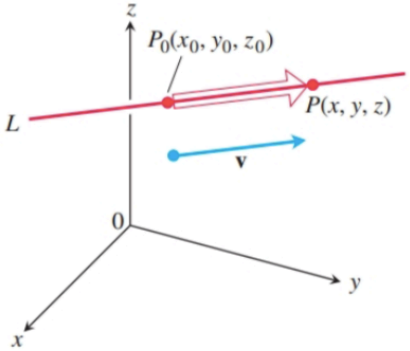
11.5 Lines and Planes in Space

Equations of a line in space

In space a line is determined by **a point** on the line and a **vector giving the direction** of the line (**a vector parallel to the line**).

Suppose L is a line in space passing through a point $P_0(x_0, y_0, z_0)$ and parallel to a vector $\mathbf{v} = v_1\,\mathbf{i} + v_2\,\mathbf{j} + v_3\,\mathbf{k}$ and $P(x, y, z)$ is any point on L.



Let $\overrightarrow{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\overrightarrow{OP_0} = \mathbf{r_0} = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$.

The line L parallel to the vector

$\Leftrightarrow$ $\overrightarrow{P_0P}$ parallel to the vector $\mathbf{v}$ $\Leftrightarrow$ $\overrightarrow{P_0P} = t\mathbf{v}$, $t \in \mathbb{R}$

$\Leftrightarrow$ $\overrightarrow{OP} - \overrightarrow{OP_0} = t\mathbf{v}$, $t \in \mathbb{R}$ $\Leftrightarrow$ $\mathbf{r} - \mathbf{r_0} = t\mathbf{v}$, $t \in \mathbb{R}$.

Equations of the line L in space

The line in space through a point $P_0(x_0, y_0, z_0)$ and parallel to a vector $\mathbf{v} = v_1\,\mathbf{i} + v_2\,\mathbf{j} + v_3\,\mathbf{k}$ has

(1) Vector equation of L: $\mathbf{r} - \mathbf{r_0} = t\mathbf{v}$, $t \in \mathbb{R}$.

That is $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle v_1, v_2, v_3 \rangle$, $t \in \mathbb{R}$.

If 2 points A and B are given, take $\mathbf{v} = \overrightarrow{AB}$.

(2) Parametric equations for L:

$$\left.\begin{aligned} x &= x_0 + t\,v_1 \\ y &= y_0 + t\,v_2 \\ z &= z_0 + t\,v_3 \end{aligned}\right\}, \quad t \in \mathrm{R} \quad (t \text{ is the parameter})$$

(3) The symmetric form for L: $\dfrac{x - x_0}{v_1} = \dfrac{y - y_0}{v_2} = \dfrac{z - z_0}{v_3}$ $(v_1 \neq 0,\ v_2 \neq 0 \text{ and } v_3 \neq 0)$

Example 1:
Find parametric equations for the line through $(-2, 0, 4)$ parallel to $\mathbf{v} = 2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$.

Solution:
Parametric equations for the line are $x = -2 + 2t$, $y = 4t$, $z = 4 - 2t$, $t \in \mathbb{R}$.

Example 2:
Find parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Solution:
$\overrightarrow{PQ} = (1 - (-3)\mathbf{i} + (-1 - 2)\mathbf{j} + \big(4 - (-3)\big)\mathbf{k} = 4\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$ is a vector parallel to the line. With $P(-3, 2, -3)$, give parametric equations $x = -3 + 4t$, $y = 2 - 3t$, $z = -3 + 7t$, $t \in \mathbb{R}$.
We could have chosen $Q(1, -1, 4)$ as the "base point" and written

$$x = 1 + 4t,\ y = -1 - 3t,\ z = 4 + 7t,\ t \in \mathbb{R}.$$

These equations serve as well as the first; they simply place you at a different point on the line for a given value of t.

Example 3:
Parametrize the line segment joining the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Solution:
From Example 2, Parametric equations for the line through P and Q,

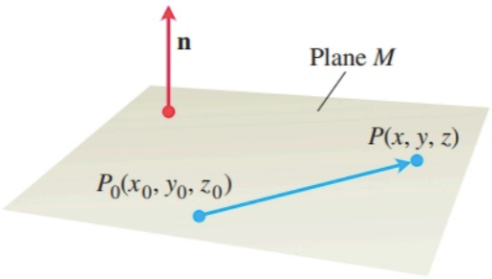
$$x = -3 + 4t,\ y = 2 - 3t,\ z = -3 + 7t,\ \ t \in \mathbb{R}.$$

At $P(-3, 2, -3)$, $-3 + 4t = -3 \Rightarrow t = 0$.
At $Q(1, -1, 4)$, $-3 + 4t = 1 \Rightarrow t = 1$.

Parametric equations for the line segment joining P and Q are

$$x = -3 + 4t,\ y = 2 - 3t,\ z = -3 + 7t,\ \ 0 \le t \le 1.$$

An Equation for a plane in space



In space a plane is determined by **a point** in the plane and a **vector normal** (perpendicular) to the plane.

Suppose M is a plane in space passes through a point $P_0(x_0, y_0, z_0)$ and is normal to the nonzero vector $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$.

Let $P(x, y, z)$ be any point on M.

Then $\overrightarrow{P_0P}$ is orthogonal to $\mathbf{n}$ $\Rightarrow$ $\overrightarrow{P_0P} \cdot \mathbf{n} = 0$

$$\Rightarrow [(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}] \cdot (a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) = 0$$

$$\Rightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\Rightarrow ax + by + cz = d \text{ where } d = ax_0 + by_0 + cz_0.$$

Equation for plane

The plane $P_0(x_0, y_0, z_0)$ and normal parallel to a vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has

(1) vector equation: $\overrightarrow{P_0P} \cdot \mathbf{n} = 0$.

(2) component equation: $\boldsymbol{A(x - x_0) + B(y - y_0) + C(z - z_0) = D}$ **.**

(3) component equation simplified: $\boldsymbol{Ax + By + Cz = D}$ **where** $\boldsymbol{D = Ax_0 + By_0 + Cz_0}$**.**

Remark:
Take note that, for any constants a and b, $x = a$, $y = b$ and $a\,x + b\,y = c$ are equations of planes in space. Example: $x = 0$, $z - 5 = 0$, $x - 2\,y = 0$, $3x + 2\,z = 1$ and $x - 3\,y + 8\,z = 1$ are equations of planes in space.

Example 4:
Find an equation for the plane through $P_0(-3, 0, 7)$ perpendicular to $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

Solution:
The component equation is $5(x - (-3)) + 2(y - 0) + (-1)(z - 7) = 0$.
Simplifying, we obtain $5x + 2y - z = -22$.

Example 5:

Find an equation for the plane through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

Solution:

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$$ is a vector normal to the plane through A, B and C.

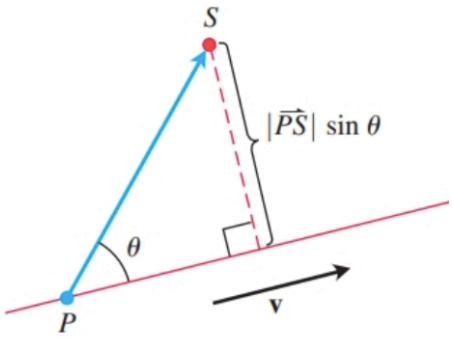
The equation of the desired plane is $3x + 2y + 6z = 3(0) + 2(0) + 6(1) = 6$.

The distance from a point to a line

The distance from a point S to a line through P parallel to $\mathbf{v}$ is given by $d = \dfrac{\left| \overrightarrow{PS} \times \mathbf{v} \right|}{\left| \mathbf{v} \right|}$.

The distance from S to the line through P parallel to $\mathbf{v}$ is $\left| \overrightarrow{PS} \right| \sin\theta$, where θ is the angle between $\overrightarrow{PS}$ and $\mathbf{v}$.

$$d = \left| \overrightarrow{PS} \right| |\sin\theta| = \frac{\left| \overrightarrow{PS} \times \mathbf{v} \right|}{\left| \mathbf{v} \right|}.$$



Example 6:

Find the distance from the point $S(1, 1, 5)$ to the line L:

$$x = 1 + t,\ y = 3 - t,\ z = 2t,\ t \in \mathbb{R}.$$

Solution:

From the equations for L that L passes through $P(1, 3, 0)$ parallel to $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

$\overrightarrow{PS} = (1-1)\mathbf{i} + (1-3)\mathbf{j} + (5-0)\mathbf{k} = -2\mathbf{j} + 5\mathbf{k}$ and

$$\overrightarrow{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = \mathbf{i} + 5\mathbf{j} + 2\mathbf{k}.$$

The distance from S to the line L is

$$d = \frac{\left| \overrightarrow{PS} \times \mathbf{v} \right|}{\left| \mathbf{v} \right|} = \frac{\sqrt{1^2 + 5^2 + 2^2}}{\sqrt{1^2 + (-1)^2 + 2^2}} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}.$$

Example 7:

Find the parametric equations for the line L through the points $P(3, 1, -2)$ and $Q(0, 7, -4)$. Find the distance from the point $S(5, 1, 0)$ to the line L. Where does the line intersect the xy-plane?

Solution:

$\overrightarrow{PQ} = -3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$ is a vector parallel to the line L.

The parametric equations for the line L are $\begin{aligned} x &= 3 - 3t \\ y &= 1 + 6t \\ z &= -2 - 2t \end{aligned}$, $t \in \mathbb{R}$.

$$\overrightarrow{PS} \times \overrightarrow{PQ} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 2 \\ -3 & 6 & -2 \end{vmatrix} = -12\mathbf{i} - 2\mathbf{j} + 12\mathbf{k} .$$

The distance from S to L is $d = \dfrac{\left| \overrightarrow{PS} \times \overrightarrow{PQ} \right|}{\left| \overrightarrow{PQ} \right|} = \dfrac{\sqrt{144 + 4 + 144}}{\sqrt{9 + 36 + 4}} = \dfrac{2\sqrt{73}}{7}$.

In the xy-plane, $z = 0$. $z = -2 - 2t = 0 \Rightarrow t = -1$.

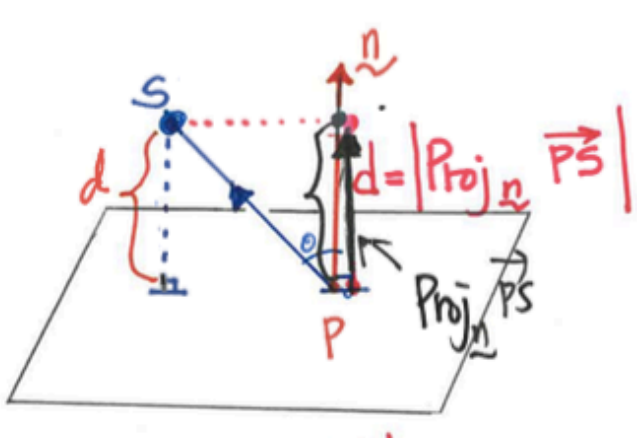
When $t = -1$, $x = 6$ and $y = -5$.

$\therefore$ The line intersects the xy-plane at $(6, -5, 0)$.

The distance from a point to a plane

If P is a point on a plane with normal $\mathbf{n}$, then the distance from any point S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$, as given in the following formula.

The distance from a point S to a plane through P normal to n is given by $d = \dfrac{\left| \overrightarrow{PS} \cdot \mathbf{n} \right|}{\left| \mathbf{n} \right|}$.



$$\begin{aligned} d &= \left| \text{Proj}_{\mathbf{n}} \overrightarrow{PS} \right| \\ &= \left| \frac{\overrightarrow{PS} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} \right| \\ &= \frac{\left| \overrightarrow{PS} \cdot \mathbf{n} \right|}{|\mathbf{n}|^2} \left| \mathbf{n} \right| \\ &= \frac{\left| \overrightarrow{PS} \cdot \mathbf{n} \right|}{|\mathbf{n}|} \end{aligned}$$

Example 8:

Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solution:

From the equation for the plane, $\mathbf{n} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$ is normal to the plane.

The points on the plane easiest to find from the plane's equation are the intercepts.

If we take P to be the y-intercept $(0, 3, 0)$, then $\overrightarrow{PS} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

Therefore, the distance from S to the plane is

$$d = \frac{\left|\overrightarrow{PS}\cdot\mathbf{n}\right|}{\left|\mathbf{n}\right|} = \frac{\left|(\mathbf{i}-2\mathbf{j}+3\mathbf{k})\cdot(3\mathbf{i}+2\mathbf{j}+6\mathbf{k})\right|}{\sqrt{9+4+36}} = \frac{\left|3-4+18\right|}{7} = \frac{17}{7}.$$

Parallel (Orthogonal) Planes

Suppose $\mathbf{n}_1$ and $\mathbf{n}_2$ are normal to the planes M_1 and M_2, respectively.

1. **Two planes M_1 and M_2 are parallel ⇔ their normal vectors are parallel**, that is $\mathbf{n}_1 = \alpha\mathbf{n}_2$ **for some** $\alpha \in \mathbb{R}$ **.**
2. **Two planes M_1 and M_2 are orthogonal ⇔ their normal vectors are orthogonal**, that is $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$.

Example 9:

Suppose the planes M_1: $2x - 3y + z = 1$ and M_2: $ax + 12y - 4z = 5$ be two planes in spaces.

(1) Find the value of a if (a) M_1 and M_2 are parallel;

(b) M_1 and M_2 are orthogonal.

(2) Find the equation for the plane through the point $P(2, 1, 3)$ and

(a) parallel to the plane M_1;

(b) perpendicular to the line $x = 3 - t$, $y = -2$, $z = 5t$, $t \in \mathbb{R}$.

Solution:

1. (a) $\mathbf{n}_1 = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ is a vector normal to M_1 and $\mathbf{n}_2 = a\mathbf{i} + 12\,\mathbf{j} - 4\mathbf{k}$ is a vector normal to M_2. $\mathbf{n}_2 = -4\left(-\frac{a}{4}\mathbf{i} - 3\mathbf{j} + \mathbf{k}\right) = -4\,\mathbf{n}_1$ when $-\frac{a}{4} = 2$.

So, M_1 and M_2 are parallel when $a = -8$.

(b) M_1 and M_2 are orthogonal ⇔ $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$ ⇔ $2a - 3(12) + 1(-4) = 0$ ⇔ $a = 20$.

2. (a) $\mathbf{n}_1$ is normal to M_1 and hence normal to the desired plane.

The equation of the desired plane is $2x - 3y + z = 2(2) - 3(1) + 1(3) = 4$.

(b) $\mathbf{v} = -\mathbf{i} + 5\mathbf{k}$ is a vector parallel to the given line and hence normal to the desired plane. The equation of the desired plane is $-x + 5z = -2 + 5(3)$, that is $x - 5z = 13$.

Intersections

(1) Line of intersection of two planes

Suppose $\mathbf{n}_1$ and $\mathbf{n}_2$ are normal to the planes M_1 and M_2, respectively.

If M_1 and M_2 are not parallel, they intersect in a line and

(i) **$n_1 \times n_2$ is a vector parallel to the intersection line** (Figure 1);

(ii) **the angle between M_1 and M_2 = the (acute) angle between n_1 and n_2** (Figure 2)

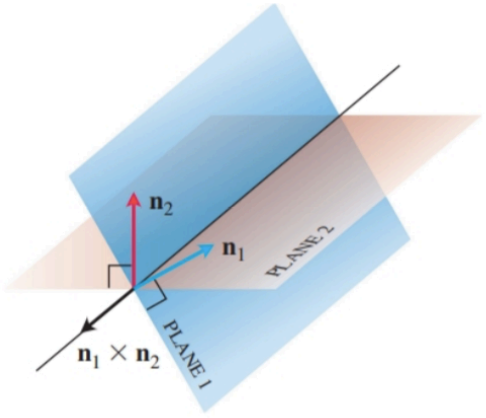


Figure 1: How the line of intersection of two planes is related to the planes' normal vectors.

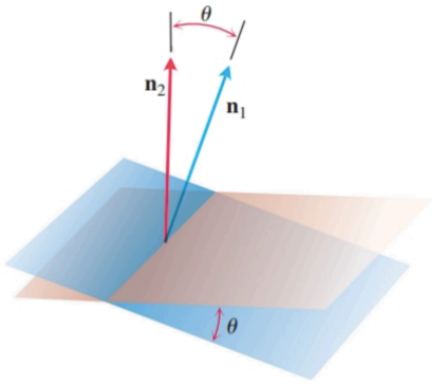


Figure 2: The angle between two planes is obtained from the angle between their normals.

Example 10: Find parametric $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ intersect.

Solution:

$\mathbf{n}_1 = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}$ is normal to the plane $3x - 6y - 2z = 15$ and

$\mathbf{n}_2 = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ is normal to the plane $2x + y - 2z = 5$.

$$\therefore \quad \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\mathbf{i} + 2\mathbf{j} + 15\mathbf{k}$$ is a vector parallel to the intersection line of the 2 planes.

To obtain a point on the line, substitute $x = 0$ into the plane equations and solve

$$-6y - 2z = 15 \quad \text{and} \quad y - 2z = 5.$$

We have $y = -\frac{10}{7}$ and $z = -\frac{45}{14}$.

So, $\left(0, -\frac{10}{7}, -\frac{45}{7}\right)$ is a point on the line.

$\therefore$ The parametric equations for the line are

$$x = 14t, \quad y = -\frac{10}{7} + 2t, \quad z = -\frac{45}{14} + 15t, \quad t \in \mathbb{R}.$$

Example 11: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution:

$\mathbf{n}_1 = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}$ is normal to the plane $3x - 6y - 2z = 15$ and

$\mathbf{n}_2 = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ is normal to the plane $2x + y - 2z = 5$.

The angle between two planes is obtained from the angle between $\mathbf{n}_1$ and $\mathbf{n}_2$.

The angle between $\mathbf{n}_1$ and $\mathbf{n}_2$ is

$$\theta = \cos^{-1}\left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1||\mathbf{n}_2|}\right) = \cos^{-1}\left(\frac{(3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} - 2\mathbf{k})}{\sqrt{9+36+4}\sqrt{4+1+4}}\right) = \cos^{-1}\left(\frac{4}{21}\right) \approx 1.38 \text{ radians.}$$

(2) Intersection of a line and a plane

Example 12: Find the point where the line $x = 8/3 + 2t,\ y = -2t,\ z = 1 + t,\ t \in \mathbb{R}$, intersects the plane $3x + 2y + 6z = 6$.

Solution:
The point $(8/3 + 2t,\ -2t,\ 1 + t)$ lies in the plane if its coordinates satisfy the equation of the plane, that is, if $3(8/3 + 2t) + 2(-2t) + 6(1 + t) = 6$

$$\Rightarrow\ 8 + 6t - 4t + 6 + 6t = 6$$
$$\Rightarrow\ 8t = -8 \ \Rightarrow\ t = -1.$$

When $t = -1$, the point of intersection is $(8/3 + 2(-1),\ -2(-1),\ 1-1) = (2/3,\ 2,\ 0)$.

Example 13:

Suppose the line L: $\frac{x-1}{2} = y = \frac{1-z}{4}$ intersects the plane S: $3x + 2z = 13$ at the point A. Find the coordinates of A and find parametric equations for the line in the plane S through A perpendicular to the line L.

Solution:
The parametric equations for the line are

$$\begin{aligned} x &= 1 + 2t, \\ y &= t, \qquad t \in R \\ z &= 1 - 4t, \end{aligned}$$

The line intersects the plane $\Leftrightarrow 3(2t + 1) + 2(1 - 4t) = 13 \ \Leftrightarrow -2t = 8 \ \Leftrightarrow\ t = -4$.

$\therefore$ The line and the plane intersect. The point of intersection is $(-7, -4,\ 17)$.

$\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ is parallel to L and $\mathbf{n} = 3\mathbf{i} + 2\mathbf{k}$ is normal to S.

A vector parallel to the required line is $\mathbf{v} \times \mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -4 \\ 3 & 0 & 2 \end{vmatrix} = 2\mathbf{i} - 16\mathbf{j} - 3\mathbf{k}$.

$\therefore$ The parametric equations of the line are $x = -7 + 2t$, $y = -4 - 16t$, $z = 17 - 3t$, $t \in \mathbb{R}$.

(3) Intersecting two lines in space

Given any two lines in space, either they are **parallel**, they **intersect** (at a point) or they are **skew** (not parallel and not intersect).

Let L_1: $\langle x, y, z \rangle = \langle x_1, y_1, z_1 \rangle + t \langle u_1, u_2, u_3 \rangle$ and

L_2: $\langle x, y, z \rangle = \langle x_2, y_2, z_2 \rangle + s \langle v_1, v_2, v_3 \rangle$ be 2 lines in space.

L_1 intersects $L_2 \Leftrightarrow$ $x_1 + t\,u_1 = x_2 + s\,v_1$, $y_1 + t\,u_2 = y_2 + s\,v_2$, $z_1 + t\,u_3 = z_2 + s\,v_3$, $s, t \in \in \mathbb{R} \Leftrightarrow$ s and t **satisfy all** the 3 equations.

Example 14:

Determine whether the following 2 lines are parallel, intersect or are skew. If they intersect, find the point of intersection.

(1) L_1: $\langle x, y, z \rangle = \langle 1, -6, 2 \rangle + t \langle 1, 2, 1 \rangle$,

L_2: $\langle x, y, z \rangle = \langle 0, 4, 1 \rangle + s \langle 2, 1, -3 \rangle$.

(2) $L_1: x - 5 = 1 - y = z - 7$ and $L_2: \dfrac{3-x}{4} = \dfrac{y-1}{2} = \dfrac{3-z}{6}$.

Solution:

(1) $\mathbf{v}_1 = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$ is parallel to L_1 and $\mathbf{v}_2 = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ is parallel to L_2.

$\mathbf{v}_1$ and $\mathbf{v}_2$ are not scalar multiple of each other

$\Rightarrow$ $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel

$\Rightarrow$ L_1 and L_2 are not parallel. ---------- (a)

L_1 and L_2 intersect $\Leftrightarrow$ $1 + t = 0 + 2s$ ---------- (1)

$-6 + 2t = 4 + s$ ---------- (2)

$2 + t = 1 - 3s$ ---------- (3).

Solve (1) & (2), $t = 7$, $s = 4$. ----------(4)

Substitute (4) into (3), $2 + t = 9 \neq 1 - 3s = -11$.

Do not exist s and t that satisfy all the equations (1) - (3).

Hence, L_1 and L_2 do not intersect. ----- (b)

Form (a) and (b), L_1 and L_2 are skew.

(2) $\mathbf{v}_1 = \mathbf{i} - \mathbf{j} + \mathbf{k}$ is parallel to L_1 and $\mathbf{v}_2 = -4\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$ is parallel to L_2.

$\mathbf{v}_1$ and $\mathbf{v}_2$ are not scalar multiple of each other

$\Rightarrow$ $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel

$\Rightarrow$ L_1 and L_2 are not parallel. ---------- (a)

L_1 and L_2 intersect $\Leftrightarrow$ $5 + t = 3 - 4s$ ---------- (1)

$1 - t = 1 + 2s$ ---------- (2)

$7 + t = 3 - 6s$ ---------- (3)

Solve (1) & (2), $s = -1$, $t = 2$. ---------- (4)

Substitute (4) into (3), $7 + t = 9 = 3 - 6s$.

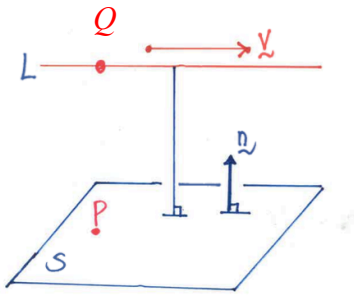
$s = -1$ and $t = 2$ satisfy the equations (1) - (3). Hence, L_1 and L_2 intersect.

When $t = 2$, $x = 5 + 2 = 7$, $y = 1 - 2 = -1$ and $z = 7 + 2 = 9$.
So, the point $(7, -1, 9)$ is the point of intersection of L_1 and L_2.

A line parallel to a plane

Suppose $\mathbf{v}$ is a vector parallel to the line L and $\mathbf{n}$ is a vector normal to the plane S.

Then the line L is parallel to the plane S if $\mathbf{v}$ and $\mathbf{n}$ are orthogonal (that is $\mathbf{v} \cdot \mathbf{n} = 0$).



Example 15:

Determine whether or not the line $L: \dfrac{x-1}{2} = \dfrac{3-y}{3} = \dfrac{z}{4}$ is parallel to the plane $2x - z = 4$.
Find the distance from the line L to the plane if they are parallel.

Solution:

$\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ is a vector parallel to the line L and $\mathbf{n} = 2\mathbf{i} - \mathbf{k}$ is vector normal to the plane $2x - z = 4$.

$\mathbf{v} \cdot \mathbf{n} = 2(2) + 4(-1) = 0 \Rightarrow \mathbf{v} \perp \mathbf{n} \Rightarrow$ the line L parallel to the plane.

The point $S(1, 3, 0)$ is on the line L and $P(2, 0, 0)$ is a point on the plane.

The distance from the line L to the plane = the distance from S to the plane

$$= \left| \frac{\overrightarrow{PS} \cdot \mathbf{n}}{|\mathbf{n}|} \right| = \left| \frac{<-1, 3, 0> \cdot <2, 0, -1>}{\sqrt{4+1}} \right| = \frac{2}{\sqrt{5}} \text{ unit.}$$